SILAS KWABLA GAH

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To:
The Selection Committee
Postdoctoral Fellow – Generative AI Researcher – 3D Defect Synthesis
Physical Science and Engineering (PSE) Division
King Abdullah University of Science and Technology (KAUST)
Thuwal 23955-6900, Kingdom of Saudi Arabia

Dear Members of the Selection Committee,

I am writing to express my enthusiastic application for the **Postdoctoral Fellow – Generative AI Researcher – 3D Defect Synthesis** position within the Physical Science and Engineering Division at King Abdullah University of Science and Technology (KAUST). As a final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (thesis defence scheduled for October 2026), my doctoral research investigates **3D geometric perception**, **inference-time foundation-model composition under extreme supervision scarcity ($\Delta\theta = 0$)**, **distribution-preserving semantic retention**, and **physics-based multi-view consensus**. My technical strengths in 3D point-cloud processing, volumetric segmentation, and uncertainty calibration directly align with your mission to synthesize geometrically and physically plausible defects in 3D CT and X-ray volumes for industrial non-destructive testing (NDT).

Addressing Data Scarcity in 3D Inspection: The Power of Distributional Retention

In AI-powered non-destructive inspection of advanced materials—such as lithium-ion battery electrodes, structural composites, and semiconductor components—critical physical defects (micro-cracks, void nucleation, porosity, inclusions, dendrite growth) are inherently rare, resulting in severe training data scarcity.

My doctoral dissertation, *Visual Perception under Low Supervision: Inference-Time Coordination of Frozen Foundation Model Representations*, directly resolves this low-data bottleneck. In my recent submission to the International Conference on Learning Representations (**ICLR 2027**), titled *Beyond Argmax: A Mechanistic Study of Semantic Retention in Frozen Foundation-Model Composition for Generalized Few-Shot 3D Segmentation*, I discovered that standard visual pipelines suffer from **premature semantic collapse**: greedy argmax selection permanently discards subtle, low-probability class hypotheses before downstream spatial reasoning can occur. In controlled experiments on 312 3D scenes ($\sim$50M points) on ScanNet200 and ScanNet++, I demonstrated that retaining compact top-k candidate distributions enables downstream geometric consensus to recover rare classes, boosting harmonic mean by +6.40 points. For 3D defect synthesis, preserving distributional alternatives across spatial voxels is essential for generating subtle morphological gradations that reflect genuine material physics.

3D Geometric Lifting, Volumetric Consensus, and Physics-Aware Supervision

My experimental background connects directly to the core requirements of 3D volumetric inspection:

- **3D Point-Cloud Segmentation & Coordinate Transforms (ACCV 2026, Paper #482):** I engineered 3D geometric lifting architectures that project camera-pose rays (K, T) to align dense 3D representations with 2D promptable segmenters (SAM 2.1) via multi-view consensus. This experience translates immediately to multi-angle X-ray projections, ray-driven forward projection, and aligning multi-scale CT volumes with downstream detection heads (YOLO, 3D U-Net).

- **Intrinsic Multi-View Process Supervision (SACAIR 2026 Submission):** I developed an intrinsic process reward that exploits multi-view physical agreement to supervise multimodal reasoning models (**Qwen2.5-VL-7B**) using online **GRPO/VERL** on **vLLM**. Through counterfactual stress-testing across 1,000+ rollouts, we eliminated reward-hacking without requiring human-labeled annotations. In defect synthesis, multi-view geometric and attenuation consistency provides an un-hackable physics prior to guide diffusion-based defect insertion.
- **Medical & Anomaly Boundary Segmentation (Springer CCIS 1841):** I co-authored peer-reviewed research developing deep encoder-decoder architectures for automated lesion boundary extraction, providing hands-on experience in fine-grained morphological segmentation.

Reliability Calibration for Safety-Critical Industrial Inspection

In structural integrity and battery safety, false negatives on defect detection can lead to catastrophic failures. To ensure trust, generative defect synthesis must produce calibrated data that does not introduce synthetic artifacts or distort downstream confidence.

In research in preparation for *IEEE TPAMI*, I formulated **Reliability-Calibrated Adaptive Semantic Arbitration (RCASA)**. RCASA introduces formal probabilistic calibration (Platt scaling, Expected Calibration Error minimization, Bernoulli uncertainty estimation) to decouple candidate confidence from contextual reliability. Proving class-wise bounded updates and probability simplex preservation, RCASA ensures that models evaluate competing physical hypotheses with mathematically certified calibration. At KAUST, I plan to leverage these calibration mechanisms to benchmark sim-to-real transfer and verify that models trained on synthesized 3D defects generalize reliably to real-world industrial CT scans.

HPC Execution and Collaborative Fit at KAUST

I bring rigorous engineering skills in **PyTorch**, **CUDA/Triton** fundamentals, and memory-bounded evaluation pipelines. I have designed streaming architectures utilizing float16 caching and chunked processing to evaluate dense 3D spatial representations on memory-constrained GPUs. I am eager to leverage KAUST's world-class computing facilities (the Shaheen supercomputing ecosystem) to train large-scale 3D diffusion and implicit neural field models for volumetric defect synthesis.

Having served as a university lecturer at Accra Technical University and founded a software consultancy, I combine deep research autonomy with proven leadership and communication skills. I would be honored to contribute to the Physical Science and Engineering Division at KAUST and push the frontiers of generative AI for material science and industrial inspection. Thank you for your consideration.

Sincerely,

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